

Solve One-Step Addition and Subtraction Equations

Lesson 7-2

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$$x + 5 = 12 \text{ — both sides equal 12 when } x = 7$$

Equation

A math sentence with an equal sign showing both sides are the same.

Addition undoes subtraction: if $x - 3 = 10$, add 3 to get $x = 13$

Inverse operation

Two math actions that undo each other, like $\times$ and $\div$.

$$x + 5 = 12 \rightarrow \text{subtract 5 from both sides} \rightarrow x = 7 \text{ (} x \text{ is isolated)}$$

Isolate

To get the letter by itself on one side.

$$x = 7 \text{ is the solution to } x + 5 = 12 \text{ because } 7 + 5 = 12$$

✓

Solution

The number that makes the equation true.

Key Ideas & Notes

- Detective Chen found a coded message: 'The suspect's locker number plus 23 equals 58.' She needs to figure out the locker number to crack the case.
- The equation is $n + 23 = 58$.
- What is n ?
- Use the balance scale to solve $n + 23 = 58$. What must you do to both sides to isolate n ?

Think About It

- What operation connects the locker number and 23?
- What does n represent in the equation?
- How could you undo the addition to find n ?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Detective Chen's clue is $n + 23 = 58$, where n is a locker number. Why can't she just 'know' n by looking, and what could she do to both sides to find it?

Level 1 support

Start here: I can undo the addition to get the variable by itself.

The variable n is added to ____ , so to find n I can ____ .

La variable n está sumada a ____ , así que para hallar n puedo ____ .

I would use ____ to undo the ____ .

Usaría ____ para deshacer la ____ .

Word bank: equation inverse operation isolate solution

Listen for: Students recognize 23 is added to n and that subtracting 23 (the inverse operation) from both sides will isolate n .

Level 2

How would your plan change if the clue were $n - 23 = 58$ instead? Justify which inverse operation you would use.

- For $n - 23 = 58$ I would ____ instead because ____ .
- The inverse operation changes because ____ .

EXPLORE

What inverse operation did you use to solve $n + 23 = 58$, and why must you do it to BOTH sides of the balance?

Level 1 support

Start here: I do the same inverse operation to both sides to keep it balanced.

I used ____ to undo ____, so the variable equals ____.

Usé ____ para deshacer ____, así que la variable es igual a ____.

I do it to both sides because ____.

Lo hago en ambos lados porque ____.

Word bank: (inverse operation) (isolate) (solution) (equation)

Listen for: Students subtract 23 from both sides to get $n = 35$ and explain that doing the same to both sides keeps the equation balanced and the solution true.

Level 2

How can you check that $n = 35$ is the solution? Show why checking matters even when your steps look correct.

- I can check by ____, which gives ____.
- Checking matters because ____.

CONNECT

Where might solving a one-step equation help you find a missing amount in real life?

Level 1 support

Start here: A one-step equation finds an unknown when I know a total and a part.

A one-step equation would help me find ____ when I know ____.

Una ecuación de un paso me ayudaría a hallar ____ cuando sé ____.

I would use the inverse operation ____ to solve it.

Usaría la operación inversa ____ para resolverla.

Word bank: **equation** **inverse operation** **isolate** **solution**

Listen for: *Students give a realistic context (money saved, distance left, missing count) and connect it to isolating a variable using an inverse operation.*

Level 2

Critique: 'Adding the same number to both sides changes the answer.' Use the idea of keeping the equation balanced to explain why this is false.

- This is false because ____.
- Doing the same operation to both sides keeps ____.

Guided Examples

Example 1

Solve: $x + 9 = 15$

Solution: Subtract 9 from both sides: $x = 15 - 9 = 6$. Check: $6 + 9 = 15$ ✓

Answer: A. $x = 6$

Example 2

Solve: $y - 7 = 20$

Solution: Add 7 to both sides: $y = 20 + 7 = 27$. Check: $27 - 7 = 20$ ✓

Answer: A. $y = 27$

Example 3

Solve: $n + 15 = 42$

Solution: Subtract 15 from both sides: $n = 42 - 15 = 27$. Check: $27 + 15 = 42$ ✓

Answer: A. $n = 27$

Write About the Math

The Writing Revolution

I can explain my steps using the words equation, inverse operation, isolate, and solution.

1. Kernel Sentence subject + verb

Model: Equation is a math sentence with an equal sign showing both sides are the same.
Ecuación es una oración matemática con un signo igual que muestra que ambos lados son iguales.

Write a kernel sentence about equation. Use a subject and a verb.

Escribe una oración base sobre ecuación. Usa un sujeto y un verbo.

2. Sentence Expansion because • but • so

Kernel: Equation matters in math
Ecuación importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Equation matters in math because ____.
Ecuación importa en matemáticas porque ____.

but
pero

Equation matters in math, but ____.
Ecuación importa en matemáticas, pero ____.

so
entonces

Equation matters in math, so ____.
Ecuación importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about equation.

Di un hecho verdadero sobre equation.

Equation ____.

Question *Pregunta*

Ask a question about equation.

Haz una pregunta sobre equation.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about equation.

Muestra entusiasmo sobre equation.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with equation.

Dile a un compañero qué hacer con equation.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

The variable n is added to ____ , **so to find n I can** ____ .

La variable n está sumada a ____ , *así que para hallar n puedo* ____ .

I would use ____ **to undo the** ____ .

Usaría ____ *para deshacer la* ____ .

I used ____ **to undo** ____ , **so the variable equals** ____ .

Usé ____ *para deshacer* ____ , *así que la variable es igual a* ____ .

Try It

Solve on your own. Check the answer key when you are done.

1. Solve: $m - 11 = 25$

- A. $m = 36$
- B. $m = 14$
- C. $m = 25$
- D. $m = 275$

Show your work:

2. A shirt costs \$ d . After a \$15 discount, it costs \$38. What is d ?

- A. \$53
- B. \$23
- C. \$38
- D. \$45

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A detective solved $x + 25 = 60$ and got $x = 35$. Another detective solved $y - 25 = 60$ and got $y = 35$. Are both correct? Explain using inverse operations and check each answer.

Sentence starter: For $x + 25 = 60$, the inverse operation is ____, so $x =$ ____ . Check: ____ . For $y - 25 = 60$, the inverse operation is ____, so $y =$ ____ . Check: ____ . Therefore, ____ .

Show your work:

Reflect — Exit Ticket

Solve: $p + 2.8 = 9.1$

- A. $p = 6.3$
- B. $p = 11.9$
- C. $p = 6.8$
- D. $p = 3.25$

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. $m = 36$ — *Add 11 to both sides: $m = 25 + 11 = 36$. Check: $36 - 11 = 25$ ✓*
2. **Try It 2:** A. $\$53 - d - 15 = 38 \rightarrow d = 38 + 15 = 53$.
3. **Exit Ticket:** A. $p = 6.3$ — *Subtract 2.8 from both sides: $p = 9.1 - 2.8 = 6.3$. Check: $6.3 + 2.8 = 9.1$ ✓*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Equation is a math sentence with an equal sign showing both sides are the same.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).